

July 28, 2022



# ASPEN AEROGELS

Q2 2022 FINANCIAL RESULTS CALL

# Disclaimer on Forward Looking Statements

This presentation contains statements that constitute “forward-looking statements” under the Private Securities Litigation Reform Act of 1995. The words “anticipate,” “believe,” “could,” “estimate,” “expect,” “guidance,” “intend,” “may,” “plan,” “potential,” “predict,” “project,” “should,” “target,” “will,” “would,” “outlook” and similar expressions are intended to identify forward-looking statements, although not all forward-looking statements contain these identifying words. All statements other than statements of historical facts contained in this presentation, including statements regarding our short-term and long-term growth strategies and expectations about the growth of our business; 2023, 2024, and 2025 revenue targets; future revenue opportunities through this decade; efforts to develop and commercialize our products; market opportunities in the electric vehicle market regarding thermal barriers as well as battery materials; market opportunities in the sustainable building materials and the energy infrastructure markets; our expectations regarding future operations, future financial position, future capacity, future revenue and projected costs; our expectations about global electric vehicle sales as well as the potential revenue from the sale of our products per electric vehicle; the revenue opportunity estimates from the major US automotive OEM ;Investment estimates for Plant II or in R&D; capital expenditure programs, annual revenue capacity with Phase I and Phase II of Plant 2, gross profit and EBITDA estimates with full manufacturing capacity; our expectations about revenue predictability; estimates of Awarded Programs in 2023 and 2025; estimates of product mixes in 2023 and 2025; our expectations about customers and automotive industry’s approach towards addressing thermal runaway; our prospects and plans; objectives of management; and expected market growth are forward-looking statements. These statements involve known and unknown risks, uncertainties and other important factors that may cause our actual results, performance or achievements to be materially different from any future results, performance or achievements expressed or implied by the forward-looking statements. These forward-looking statements are only predictions, and we may not actually achieve the plans, intentions or expectations disclosed in our forward-looking statements, so you should not rely on our forward-looking statements. Actual results or events could differ materially from the plans, intentions and expectations disclosed in the forward-looking statements we make. We have based these forward-looking statements largely on our current expectations and projections about future events and trends that we believe may affect our business, financial condition and operating results. The Company’s future financial and operating results, as well as any forward-looking statements, are subject to inherent risks and uncertainties, such as those described in its Annual Report on Form 10-K for the year ended December 31, 2021, as well as its subsequent annual, quarterly and current reports filed with the U.S. Securities and Exchange Commission. Such risks and uncertainties include those related to: an inability to create new product, partnership and market opportunities; any sustained downturn in the energy industry or energy prices; any sustained downturn in the petrochemical, refinery, subsea, LNG, U.S., North American, Asian, European, South American, Middle Eastern, building materials, electric vehicle or other market due to the coronavirus pandemic, COVID-19 or any other factor; any failure to sustain project-based demand in the subsea, LNG, on-shore or other markets; the right of EV thermal barrier customers to cancel contracts with Aspen at any time and without penalty; any costs, expenses, or investments incurred by Aspen in excess of projections used to develop pricing under the contracts with EV thermal barrier customers; any failure of Aspen or PyroThin® to meet contractual specifications and requirements under contracts with EV thermal barrier customers; Aspen’s inability to create new product, customer or market opportunities, including for PyroThin® or for new products developed from Aspen’s aerogel technology; any disruption or inability to achieve expected capacity levels in any of our three production lines or at Plant II including due to the coronavirus pandemic, COVID-19 or any other factor, any inability to expand manufacturing capacity Plant II; Aspen’s Pyrothin® product or another product’s ability to solve the customer problem towards which it is targeted; the failure to receive all regulatory or other approvals required to operate, maintain or expand our facilities; any failure of demand for Aspen’s products; any failure to achieve expected price increases or average selling prices for Aspen’s products; any significant increase in the cost of raw materials, utilities or any other manufacturing consumable; the failure to mitigate the impact of any significant increase in the cost of raw materials, utilities or other manufacturing consumable; shortages of raw materials, utilities or any other manufacturing consumable due to the coronavirus pandemic, COVID-19 or any other factor; the failure to generate sufficient operating cash flow or to obtain significant additional capital to pursue Aspen’s strategy; the failure of Aspen’s products to become widely adopted; the competition Aspen faces in its business; any failure to enforce any of Aspen’s patents; any failure to protect or expand Aspen’s aerogel technology platform; any future finding of invalidity of any patent in any jurisdiction; any failure to generate sufficient operating cash flow or to obtain sufficient additional capital to continue to pursue Aspen’s new business, technology, patent enforcement, or patent defense strategy; any failure of Aspen’s products to meet applicable specifications and other performance, safety, technical and delivery requirements; the general economic conditions and cyclical demands in the markets that Aspen serves; the economic, operational and political risks associated with sales and expansion of operations in foreign countries; the loss of any direct customer, including distributors, contractors and OEMs; compliance with health and safety laws and regulations; and the maintenance and development of distribution channels.

# Disclaimer on Forward Looking Statements

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The industry and market data contained in this presentation is based either on our management's own estimates or on independent industry publications, reports by market research firms or other publicized independent sources. Although the Company believes these sources are reliable, it has not independently verified the information and cannot guarantee its accuracy and completeness, as industry and market data are subject to change and cannot always be verified with complete certainty due to limits on the availability and reliability of raw data, the voluntary nature of the data gathering process and other limitations and uncertainties inherent in any statistical survey or market shares. Accordingly, you should be aware that the industry and market data contained in this presentation, and estimates and beliefs based on such data, may not be reliable. Unless otherwise indicated, all information contained in this presentation concerning our industry in general or any segment thereof, including information regarding our general expectations and market opportunity, is based on management's estimates using internal data, data from industry related publications, consumer research and marketing studies and other externally obtained data. Certain financial or other targets are based on management estimates, currently available information and assumptions that may change. Accordingly, there can be no assurance that we will achieve our financial targets or other expectations. The expectations are inherently subject to significant economic, competitive and other uncertainties and contingencies, many of which are beyond the control of management. Actual results may vary materially based on a number of factors. For example, The Company's estimates in this presentation are based on assumptions about the following factors: (i) adoption rates of EV and our customers' market penetration, (ii) the aggregate pricing of our products, and (iii) the average amount of our products incorporated in each vehicle. Due to the rapid pace of technical development in lithium batteries, the Company's assumptions may evolve in the future due to various external factors resulting in different estimates than those arrived here. There are a wide range of estimates being published, including ones that would result in lower per vehicle figures. While the Company's management believes its estimates regarding possible pricing and market size are based on reasonable assumptions and methodology, there can be no assurance that the future pricing and market size will align with the Company's estimates. Such data may be outdated and involves a number of assumptions and limitations and contains projections and estimates of the future performance of the industries in which we operate that are subject to a high degree of uncertainty. We caution you not to give undue weight to such projections, assumptions and estimates.

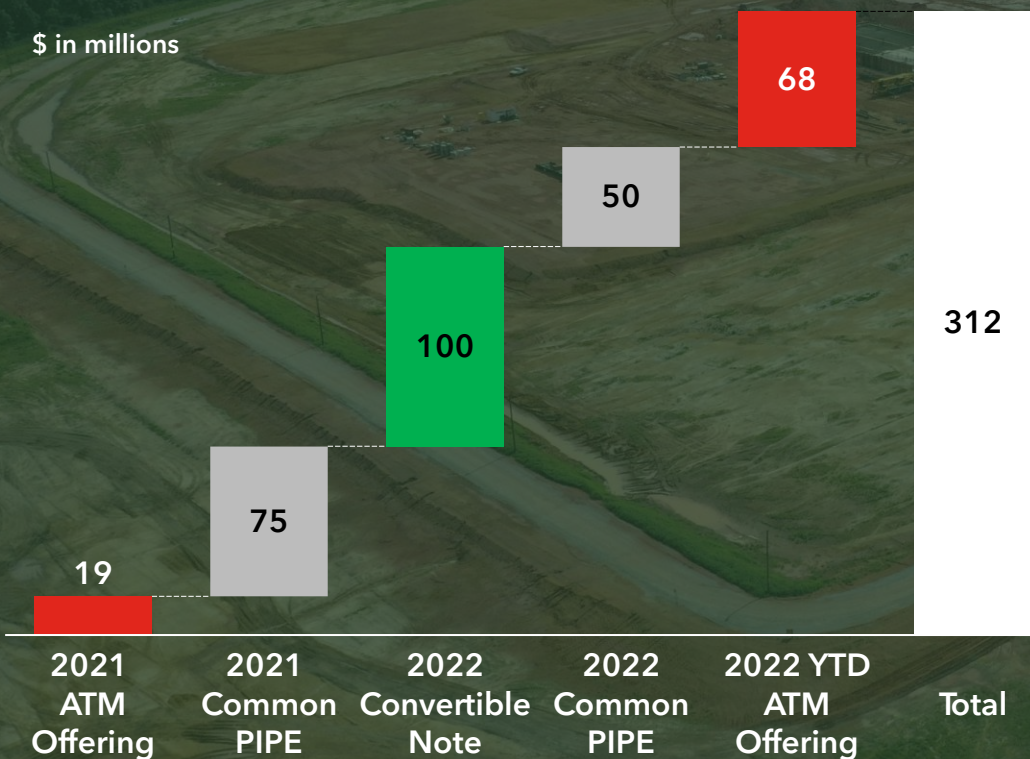


# Recent Financing Activities and Q2 Developments

Advancing a plan designed to turn Aspen Aerogels into a potential \$1.6B revenue and \$400M adjusted EBITDA business

## Recent Financing Activities

- \$312M raised since January of 2021



## Q2 and Recent Highlights

- 44% YoY Revenue Growth
- 19% QoQ Revenue Growth
- Strong Energy Industrial growth driven by maintenance demand, high energy prices and geopolitical environment
- Continued EV thermal barrier volumes growth
- Mexico EV thermal barrier assembly facility in Apodaca, Nuevo Leon has started delivering end-customer parts
- Aerogel Plant 2 on track with \$62.5M spent through Q2

## Sales Growth Drivers

|                                                                                       |                   | Quarter Over Quarter | Year Over Year |
|---------------------------------------------------------------------------------------|-------------------|----------------------|----------------|
|  | THERMAL BARRIERS  | 42%                  | 34X growth     |
|  | ENERGY INDUSTRIAL | 13%                  | 10%            |

# Q2 2022 Financial Highlights

44% YoY revenue growth driven by strong demand across all markets and higher than expected thermal barrier volumes

| \$ in millions except per share figures | Q2 2021       | Q2 2022 <sup>1</sup> | H1 2021       | H1 2022 <sup>1</sup> | 2022 Outlook <sup>1</sup>         | Commentary                                                                                                                                                                     |
|-----------------------------------------|---------------|----------------------|---------------|----------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REVENUES<br>YoY Growth                  | \$31.7<br>29% | \$45.6<br>44%        | \$59.8<br>13% | \$84.0<br>40%        | \$180.0 - \$200.0<br>48%      64% | Continued PyroThin <sup>®</sup> volume growth and energy infrastructure product demand globally.                                                                               |
| Net (Loss)                              | \$(6.7)       | \$(24.0)             | \$(12.9)      | \$(43.5)             | \$(79.8) - \$(86.8)               | Significant investment in talent, systems, facilities and tooling to enable continuous EV market growth and productivity investments aimed at driving long-term profitability. |
| Adj. EBITDA <sup>2</sup>                | \$(3.4)       | \$(18.3)             | \$(6.0)       | \$(33.0)             | \$(55.0) - \$(62.0)               |                                                                                                                                                                                |
| EPS                                     | \$(0.23)      | \$(0.68)             | \$(0.46)      | \$(1.27)             | \$(2.16) - \$(2.36)               | Investments aligned to deliver customer demand and enable supply chain resiliency.                                                                                             |
| Cap Ex                                  | \$2.4         | \$37.9               | \$3.9         | \$52.3               | \$250.0 - \$300.0                 | R&D facility upgrades, Plant II construction, PyroThin <sup>®</sup> fabrication tooling in Mexico, IT systems.                                                                 |

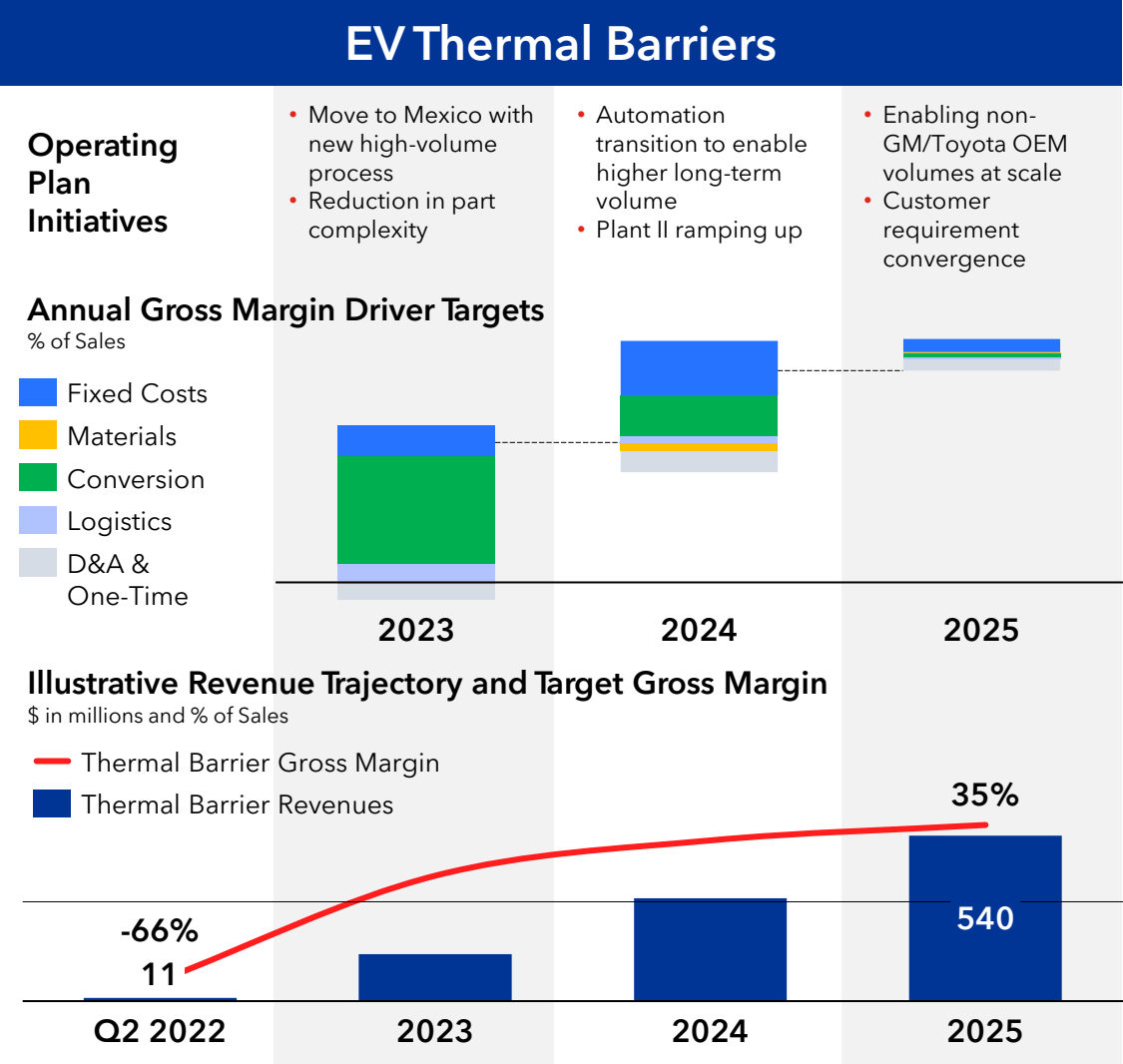
1- Unaudited financials for 2022 (and Outlook)

2- See slide 10 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented period



# Operating Plan to Drive Down the Cost of Sales

Managing capacity transitions and expansion across both segments enables the path to our target gross margins



# OEMs' Increasing Cell-to-Cell Barrier Adoption

7 of the top 8 EV OEMs use compatible cell form factors and can add cell-to-cell barriers to enhance safety

| OEM Group                 | Market Position <sup>1</sup> |                      | Battery Strategy <sup>2</sup> |                  | Passive Thermal Runaway Solutions <sup>3</sup> |               |                 |                                   | Approach Summary                                                          |
|---------------------------|------------------------------|----------------------|-------------------------------|------------------|------------------------------------------------|---------------|-----------------|-----------------------------------|---------------------------------------------------------------------------|
|                           | 2021 Global EV Share         | 2025 Global EV Share | Cell Chemistry                | Cell Form Factor | Cooling System                                 | Pack Barriers | Module Barriers | Cell to Cell Barriers             |                                                                           |
| TESLA                     | 20%                          | 20%                  | LFP / NCA                     | Cylindrical      | ✓                                              | ✓             |                 | Bonding Foams                     | Invested in current design that uses small cylindrical cells structurally |
| gm                        | 10%                          | 9%                   | NMC                           | Pouch            | ✓                                              | ✓             | ✓               | PyroThin™                         | Leading and optimizing system with a scalable platform approach           |
| STELLANTIS                | 4%                           | 9%                   | NMC                           | Prismatic        | ✓                                              | ✓             | ✓               | PyroThin® Compatible              | Battery JV assessing designs to increase protection                       |
| VW                        | 10%                          | 8%                   | NMC                           | Prismatic        | ✓                                              | ✓             |                 |                                   | Increasing focus on cell-to-cell protection for next gen platforms        |
| BYD                       | 7%                           | 6%                   | LFP                           | Prismatic        | ✓                                              | ✓             | ✓               |                                   | Increased focus on large and long prismatic cells for temp dissipation    |
| RENAULT NISSAN MITSUBISHI | 5%                           | 6%                   | NMC                           | Pouch            | ✓                                              | ✓             | ✓               |                                   | Continuously assessing new safety measures for thermal runaway            |
| Ford                      | 1%                           | 5%                   | NMC                           | Pouch            | ✓                                              | ✓             | ✓               | PyroThin™                         | Continuously assessing new safety measures for thermal runaway            |
| TOYOTA                    | 0.4%                         | 2%                   | NMC                           | Prismatic        | ✓                                              | ✓             | ✓               |                                   | Incorporating cell to cell barriers in new EV nameplates in the future    |
| All Others                | 42%                          | 35%                  | Various                       | Various          | Various                                        | Various       |                 | Generally Compatible <sup>4</sup> | Increasingly aware of thermal runaway and thermal propagation             |

# Relevant Regulatory Landscape Developments

Warning of thermal propagation and runaway is a key part of emerging global EV safety and testing standards

## EMERGING REGULATIONS AND TESTING STANDARDS AIMED AT EV SAFETY...



### UN GTR No. 20 - EV Safety (Adopted 2018)<sup>1</sup>

*"The vehicle shall provide an advance warning indication to allow egress of 5 minutes prior to the presence of a hazardous situation inside the passenger compartment caused by **thermal propagation** which is triggered by an internal short circuit leading to a **single cell thermal runaway such as fire, explosion or smoke.**"*

## ARE DRIVING COUNTRY LEVEL STANDARDS FOR THERMAL PROPAGATION

### Country Level Implementation of GTR No. 20

| Region                                                                                | Key Action Dates            | Recent or Expected Outcomes <sup>3</sup>                                                                         |
|---------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------|
|    | 2022<br>(notice & comment)  | NHTSA to propose GTR 20 as amendment to Motor Vehicle Safety Standards ("MVSS") for public comment in Sept. 2022 |
|    | 2021<br>(in force)          | Harmonized with GTR20 and "adopted them in Safety Regulations for Road Vehicles"                                 |
|    | 2022<br>(mandatory for all) | Indian Automotive Industry Standard revised "to be in line with GTR20"                                           |
|    | 2021<br>(mandatory for new) | Adopted GB38031 to "establish relevant National Regulation harmonized with GTR20."                               |
|    | 2021<br>(mandatory for new) | Rechargeable Electrical Energy Storage Systems in Korea MVSS harmonized w/ GTR20                                 |
|   | 2025<br>(harmonization)     | Russia to follow the Eurasian Economic Union; expected to follow GTR20                                           |
|  | TBD                         | Follows US in adopting GTR20 ("[Canadian] MVSS references ... [US] MVSS.")                                       |
|  | TBD                         | EU transposition proceeds with translation of the amendments in EU all languages                                 |

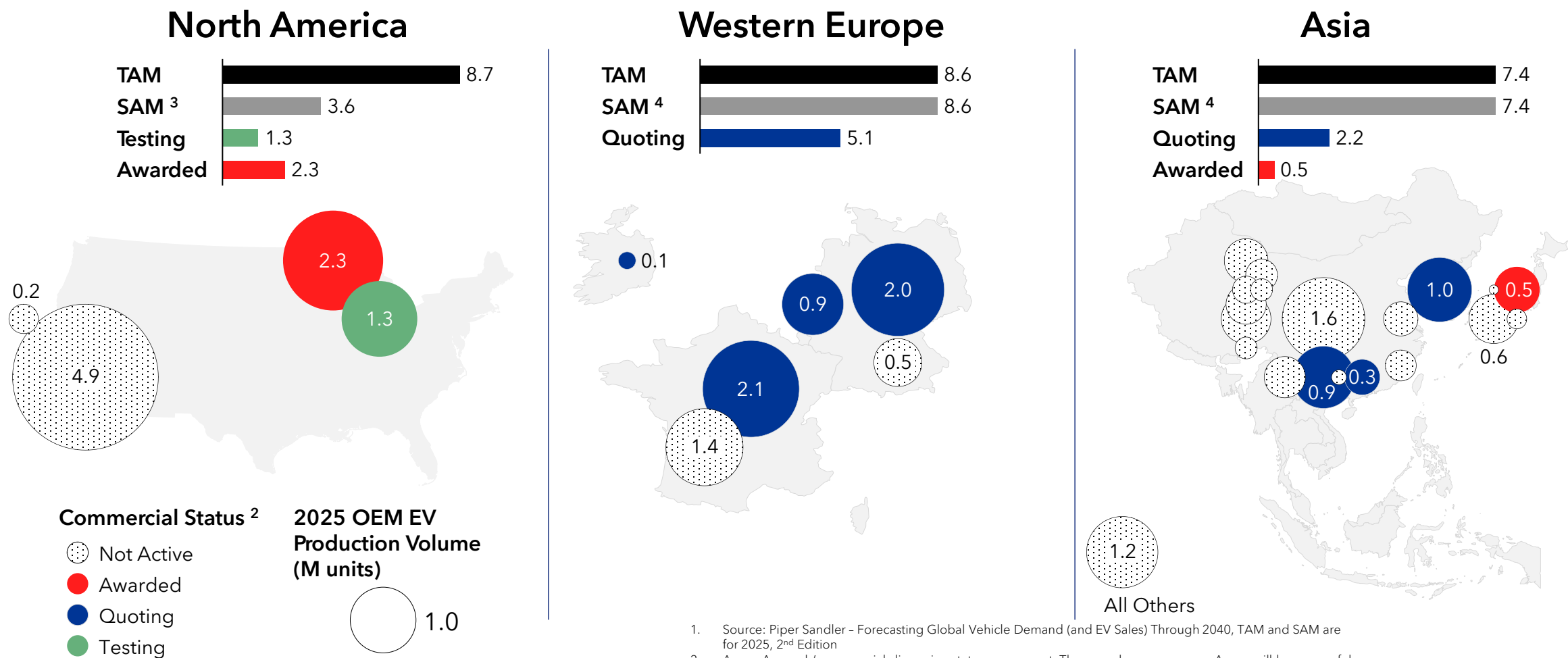
1. Source: Global Technical Regulation on the Electric Vehicle Safety (EVS), Addendum 20: Global Technical Regulation No. 20, May 3, 2018
2. Source: National Standards of the People's Republic of China - Electric Vehicles Traction Battery Safety Requirements, GB 38031-2020, May 12, 2020
3. Source: Transposition of GTR20 (Phase 1) to National Regulation - Progress Report - 23rd EVS-GTR IWG, Mar. 2022  
Recent or expected outcomes from source documents; UNECE or member state projections, not Aspen's



# EV Thermal Barrier Commercial Progress

Aspen has been awarded business, is quoting, or in discussions with several OEMs that it believes will drive EV production

## 2025 Forecasted EV Production Volume (M units) and Sourcing Status by OEM Group HQ Location <sup>1</sup>



- Source: Piper Sandler - Forecasting Global Vehicle Demand (and EV Sales) Through 2040, TAM and SAM are for 2025, 2<sup>nd</sup> Edition
- Aspen Aerogels' commercial discussion status assessment. There can be no assurance Aspen will be successful in generating new business from these discussions. In addition, there can be no assurance awarded business will result in sales
- SAM adjusted to reflect volume reduction of OEMs using cylindrical cells
- Management estimates through 2025 expect there will be no OEMs using cylindrical cells

# Our Strategy to Create Stakeholder Value

Leveraging our platform in markets with favorable dynamics while focusing on doing the right things well creates value

## WHO WE ARE



PROPRIETARY AEROGEL  
TECHNOLOGY PLATFORM™,  
PROCESSES, SCALE AND  
CRITICAL TALENT BASE

## WHERE WE PLAY

AT THE CRUX OF 4 ACCELERATING  
TRENDS REQUIRING SOLUTIONS



EV SAFETY, RELIABILITY,  
SERVICEABILITY



ENERGY DENSITY, COST,  
RELIABILITY



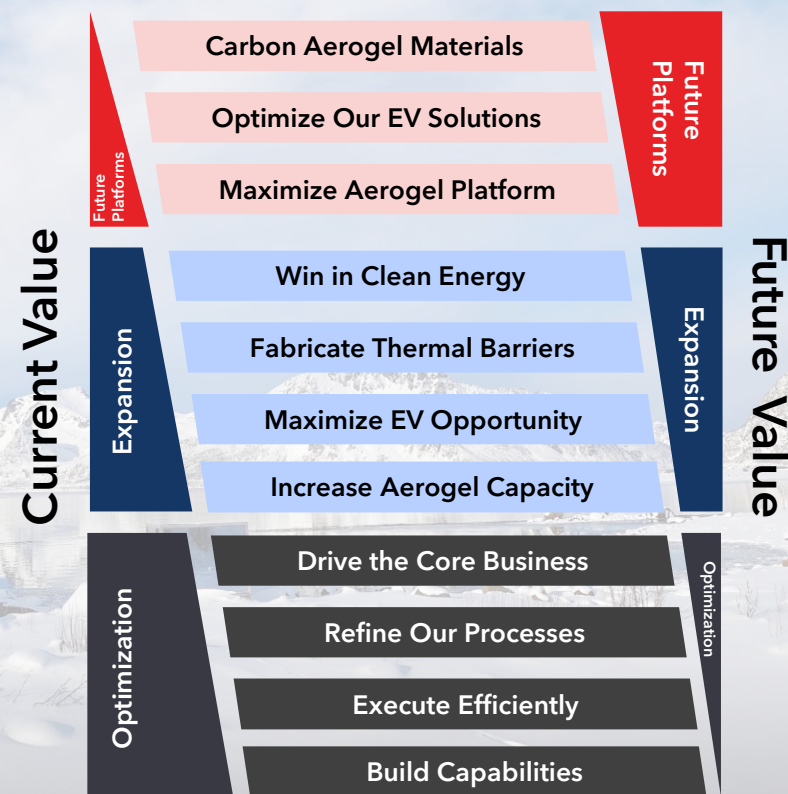
ASSET EFFICIENCY,  
RESILIENCY, SAFETY



THERMAL EFFICIENCY  
AND FIRE SAFETY

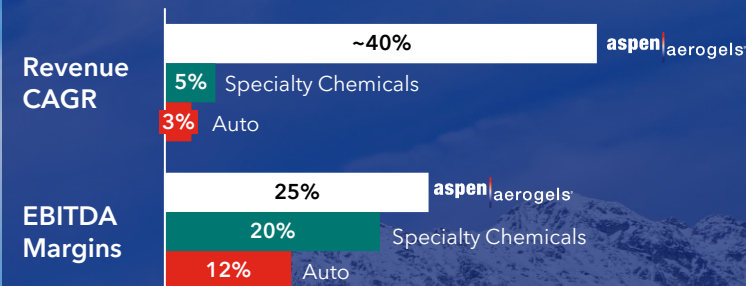
## WHAT WE DO

OPTIMIZE AND EXPAND OUR PLATFORM  
WHILE INVESTING TOWARDS THE FUTURE



## CREATES VALUE

Peers' <sup>1</sup> Revenue Growth / EBITDA Margins <sup>2</sup> side by side  
with Aspen's targeted performance at Full Capacity <sup>3</sup>



Peers' <sup>1</sup> TEV, Revenue Growth / EBITDA Margins <sup>2</sup> relative  
to Aspen's targeted performance at Full Capacity <sup>3</sup>



# Appendix

GAAP-Non-GAAP Reconciliation



# GAAP to Non-GAAP Reconciliation

| (\$ in thousands)             | Q2              |                | Q2 YTD          |                | FY 2022         |                 |
|-------------------------------|-----------------|----------------|-----------------|----------------|-----------------|-----------------|
|                               | 2022            | 2021           | 2022            | 2021           | Low             | High            |
| Net loss                      | (24,050)        | (6,669)        | (43,534)        | (12,919)       | (86,800)        | (79,800)        |
| Depreciation and Amortization | 2,032           | 2,104          | 4,161           | 4,742          | 9,700           | 9,700           |
| Stock-based Compensation      | 2,295           | 1,070          | 4,123           | 2,046          | 8,300           | 8,300           |
| Interest Expense, net         | 1,404           | 55             | 2,264           | 130            | 6,800           | 6,800           |
| <b>Adjusted EBITDA</b>        | <b>(18,319)</b> | <b>(3,440)</b> | <b>(32,986)</b> | <b>(6,001)</b> | <b>(62,000)</b> | <b>(55,000)</b> |



An aerial photograph of a dense evergreen forest covered in a thick layer of snow. A dark, winding road or path cuts through the center of the forest, curving from the bottom left towards the top right. The trees are closely packed, and their branches are heavily laden with snow, creating a textured, white landscape. The road is a dark, straight line that contrasts with the white snow and dark green of the trees.

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